

La base des bascules

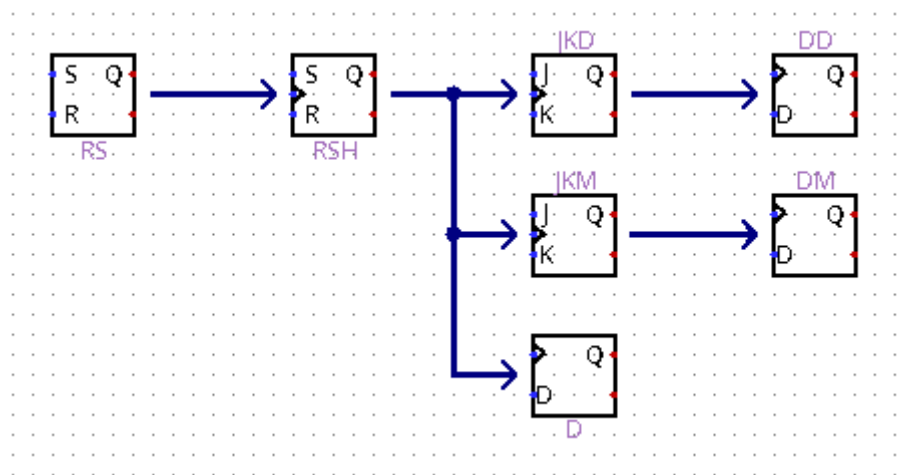
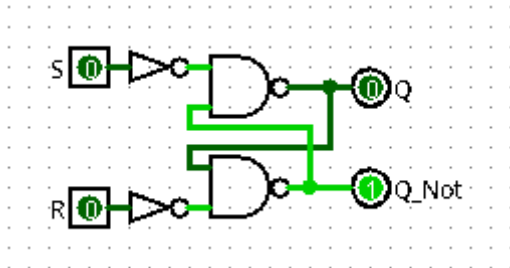


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1) Bascule RS :

1-1) NON ET : (RS1)



$$Q = Q_{n-1} \cdot \bar{R} + S$$

$$\bar{Q} = Q_{n-1} \cdot \bar{S} + R$$

S	R	Q	$\bar{Q}$	remarque
0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	1	mise à 0
1	0	1	0	mise à 1
1	1	X	X	état interdit

Algèbre de boole

$$Q = \bar{\bar{Q}} \cdot \bar{S}$$

$$\bar{Q} = \bar{Q} \cdot \bar{R}$$

$$Q = \overline{(\bar{Q} \cdot \bar{R}) \cdot S}$$

$$Q = \overline{(\bar{Q} + \bar{R}) \cdot S}$$

$$Q = \overline{(\bar{Q} + R) \cdot S}$$

$$Q = \overline{\bar{Q} \cdot \bar{S} + R \cdot \bar{S}}$$

$$Q = \overline{(\bar{Q} \cdot \bar{S}) + (R \cdot \bar{S})}$$

$$Q = \overline{(\bar{Q} \cdot \bar{S}) \cdot (R \cdot \bar{S})}$$

$$Q = \overline{(\bar{Q} + \bar{S}) \cdot (\bar{R} + \bar{S})}$$

$$Q = \overline{(Q + S) \cdot (\bar{R} + S)}$$

$$Q = \overline{(Q + S) \cdot (\bar{R} + S)}$$

$$Q = Q \cdot \bar{R} + Q \cdot S + S \cdot \bar{R} + S \cdot S$$

$$Q = Q \cdot \bar{R} + S \cdot Q + S \cdot \bar{R} + S \cdot 1$$

$$Q = Q \cdot \bar{R} + S \cdot (Q + \bar{R} + 1)$$

$$Q = Q \cdot \bar{R} + S$$

$$Q = Q_{n-1} \cdot \bar{R} + S$$

$$\bar{Q} = \overline{(\bar{Q} \cdot \bar{S}) \cdot R}$$

$$\bar{Q} = \overline{(\bar{Q} + \bar{S}) \cdot R}$$

$$\bar{Q} = \overline{(Q + S) \cdot \bar{R}}$$

$$\bar{Q} = \overline{Q \cdot \bar{R} + S \cdot \bar{R}}$$

$$\bar{Q} = \overline{(Q \cdot \bar{R}) + (S \cdot \bar{R})}$$

$$\bar{Q} = \overline{(Q \cdot \bar{R}) \cdot (S \cdot \bar{R})}$$

$$\bar{Q} = \overline{(\bar{Q} + \bar{R}) \cdot (\bar{S} + \bar{R})}$$

$$\bar{Q} = \overline{(\bar{Q} + R) \cdot (\bar{S} + R)}$$

$$\bar{Q} = \overline{(\bar{Q} + R) \cdot (\bar{S} + R)}$$

$$\bar{Q} = \bar{Q} \cdot \bar{S} + \bar{Q} \cdot R + R \cdot \bar{S} + R \cdot R$$

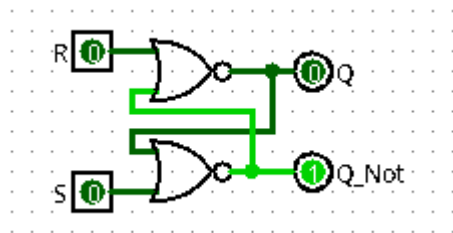
$$\bar{Q} = \bar{Q} \cdot \bar{S} + \bar{Q} \cdot R + R \cdot \bar{S} + 1 \cdot R$$

$$\bar{Q} = \bar{Q} \cdot \bar{S} + R \cdot (\bar{Q} + \bar{S} + 1)$$

$$\bar{Q} = \bar{Q} \cdot \bar{S} + R$$

$$\bar{Q} = \bar{Q}_{n-1} \cdot \bar{S} + R$$

1-2) NON OU :



$$Q = Q_{(n-1)} \cdot \bar{R} + S$$

$$\bar{Q} = \overline{Q_{(n-1)} \cdot \bar{S} + R}$$

S	R	Q	$\bar{Q}$	remarque
0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	1	mise à 0
1	0	1	0	mise à 1
1	1	X	X	état interdit

Algèbre de boole

$$Q = \overline{\bar{Q} + R}$$

$$\bar{Q} = \overline{Q + S}$$

$$\bar{Q} = \overline{\bar{Q} + R + S}$$

$$\bar{Q} = (Q \cdot \bar{R}) + S$$

$$\bar{Q} = (Q + S) \cdot (\bar{R} + S)$$

$$\bar{Q} = Q \cdot \bar{R} + Q \cdot S + S \cdot \bar{R} + S \cdot S$$

$$\bar{Q} = Q \cdot \bar{R} + Q \cdot S + S \cdot \bar{R} + S \cdot 1$$

$$\bar{Q} = S \cdot (Q + \bar{R} + 1) + Q \cdot \bar{R}$$

$$\bar{Q} = S + Q \cdot \bar{R}$$

$$\bar{Q} = \bar{S} \cdot \bar{Q} + R$$

$$\bar{Q} = \bar{Q} \cdot \bar{S} + R$$

$$\bar{Q} = Q_{(n-1)} \cdot \bar{S} + R$$

$$Q = \overline{\bar{Q} + \bar{S} + R}$$

$$Q = (\bar{Q} \cdot \bar{S}) + R$$

$$Q = (\bar{Q} + R) \cdot (\bar{S} + R)$$

$$Q = \bar{Q} \cdot \bar{S} + \bar{Q} \cdot R + R \cdot \bar{S} + R \cdot R$$

$$Q = \bar{Q} \cdot \bar{S} + \bar{Q} \cdot R + R \cdot \bar{S} + R \cdot 1$$

$$Q = R \cdot (\bar{Q} + \bar{S} + 1) + \bar{Q} \cdot \bar{S}$$

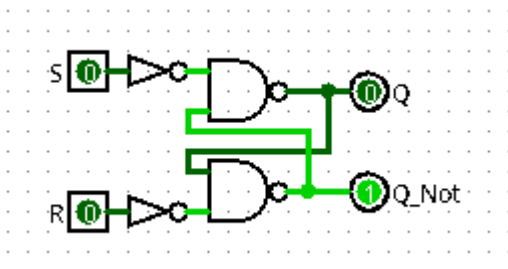
$$Q = R + \bar{Q} \cdot \bar{S}$$

$$Q = \bar{R} \cdot Q + S$$

$$Q = Q \cdot \bar{R} + S$$

$$Q = Q_{(n-1)} \cdot \bar{R} + S$$

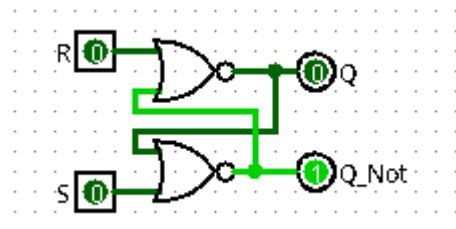
1-3) Résultat :



$$Q = Q_{n-1} \cdot \bar{R} + S$$

$$\bar{Q} = \overline{Q_{n-1} \cdot \bar{S} + R}$$

S	R	Q	$\bar{Q}$	remarque
0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	1	mise à 0
1	0	1	0	mise à 1
1	1	X	X	état interdit

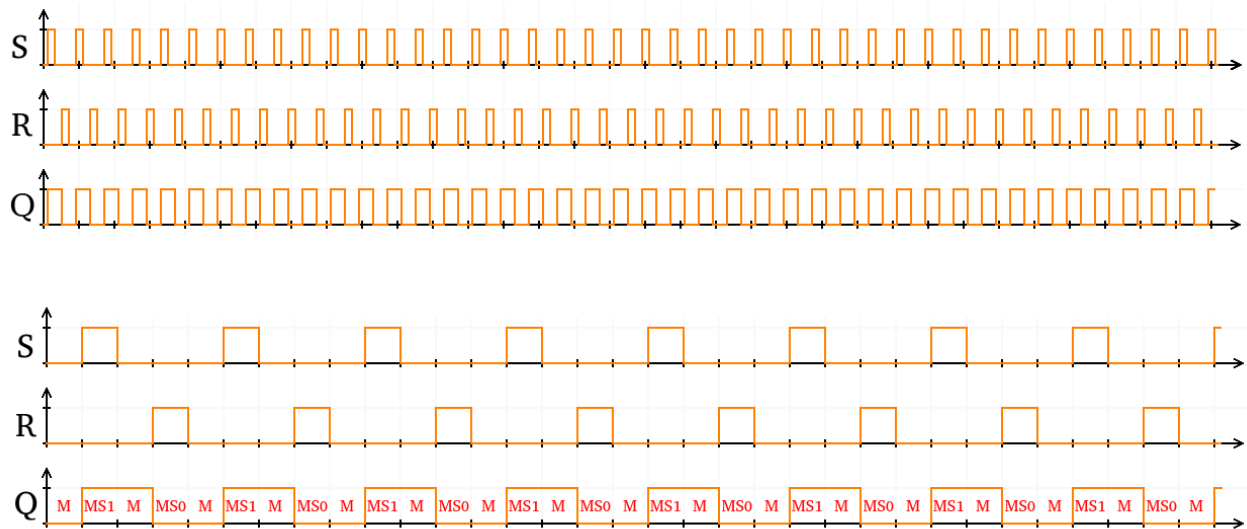


$$Q = Q_{(n-1)} \cdot \bar{R} + S$$

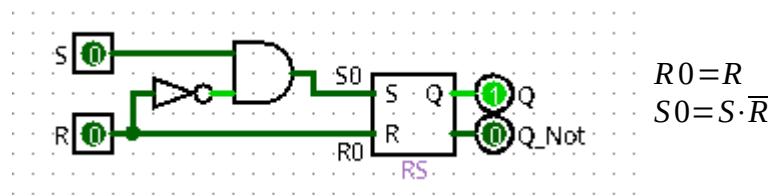
$$\bar{Q} = \overline{Q_{(n-1)} \cdot \bar{S} + R}$$

S	R	Q	$\bar{Q}$	remarque
0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	1	mise à 0
1	0	1	0	mise à 1
1	1	X	X	état interdit

1-4) diagramme de temps :

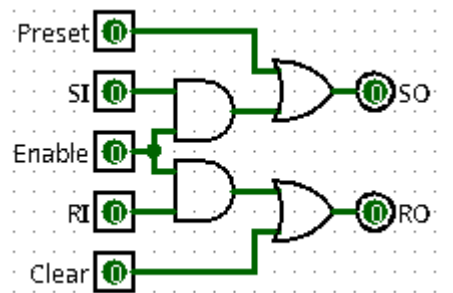


1-5) Logisim : (RSL)



2) Preset Enable Clear :

2-1) PECM : (Preset Enable Clear Main)



$$SO = P + (SI \cdot E)$$

$$RO = C + (RI \cdot E)$$

E=0 P=0 C=0

$$SO = 0$$

$$RO = 0$$

$$SO = 0 + (SI \cdot 0)$$

$$SO = 0 + 0$$

$$SO = 0$$

$$RO = 0 + (RI \cdot 0)$$

$$RO = 0 + 0$$

$$RO = 0$$

Mémoire

E=1 P=0 C=0

$$SO = SI$$

$$RO = RI$$

$$SO = 0 + (SI \cdot 1)$$

$$SO = 0 + SI$$

$$SO = SI$$

$$RO = 0 + (RI \cdot 1)$$

$$RO = 0 + RI$$

$$RO = RI$$

Bascule RS

E=0 P=1 C=0

$$SO = 1$$

$$RO = 0$$

$$SO = 1 + (SI \cdot 0)$$

$$SO = 1 + 0$$

$$SO = 1$$

$$RO = 0 + (RI \cdot 0)$$

$$RO = 0 + 0$$

$$RO = 0$$

Mise à 1

E=0 P=0 C=1

$$SO = 0$$

$$RO = 1$$

$$SO = 0 + (SI \cdot 0)$$

$$SO = 0 + 0$$

$$SO = 0$$

$$RO = 1 + (RI \cdot 0)$$

$$RO = 1 + 0$$

$$RO = 1$$

Mise à 0

E=0 P=1 C=1

$$SO = 1$$

$$RO = 1$$

$$SO = 1 + (SI \cdot 0)$$

$$SO = 1 + 0$$

$$SO = 1$$

$$RO = 1 + (RI \cdot 0)$$

$$RO = 1 + 0$$

$$RO = 1$$

Interdit

E=1 P=1 C=1

$$SO = 1$$

$$RO = 1$$

$$SO = 1 + (SI \cdot 1)$$

$$SO = 1 + SI$$

$$SO = 1$$

$$RO = 1 + (RI \cdot 1)$$

$$RO = 1 + RI$$

$$RO = 1$$

Interdit

E=1 P=0 C=1

$$SO = SI$$

$$RO = 1$$

$$SO = 0 + (SI \cdot 1)$$

$$SO = 0 + SI$$

$$SO = SI$$

$$RO = 1 + (RI \cdot 1)$$

$$RO = 1 + RI$$

$$RO = 1$$

SI=0 : Mise à 0
SI=1 : Interdit

E=1 P=1 C=0

$$SO = 1$$

$$RO = RI$$

$$SO = 1 + (SI \cdot 1)$$

$$SO = 1 + SI$$

$$SO = 1$$

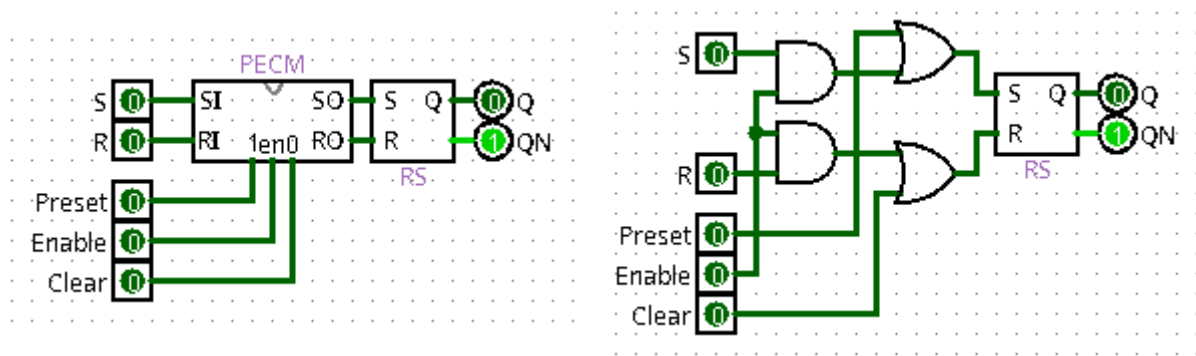
$$RO = 0 + (RI \cdot 1)$$

$$RO = 0 + RI$$

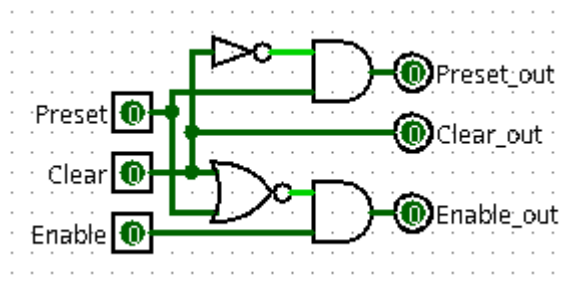
$$RO = RI$$

RI=0 : Mise à 1
RI=1 : Interdit

2-2) RSEM : (RS Enable Main)



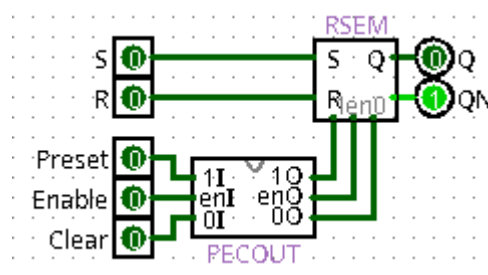
2-3) Logisim : (PECOUT)



$$\begin{aligned} \text{ClearOut} &= \text{Clear} \\ \text{PresetOut} &= \text{Preset} \cdot \overline{\text{Clear}} \\ \text{EnableOut} &= \text{Enable} \cdot (\text{Preset} + \text{Clear}) \end{aligned}$$

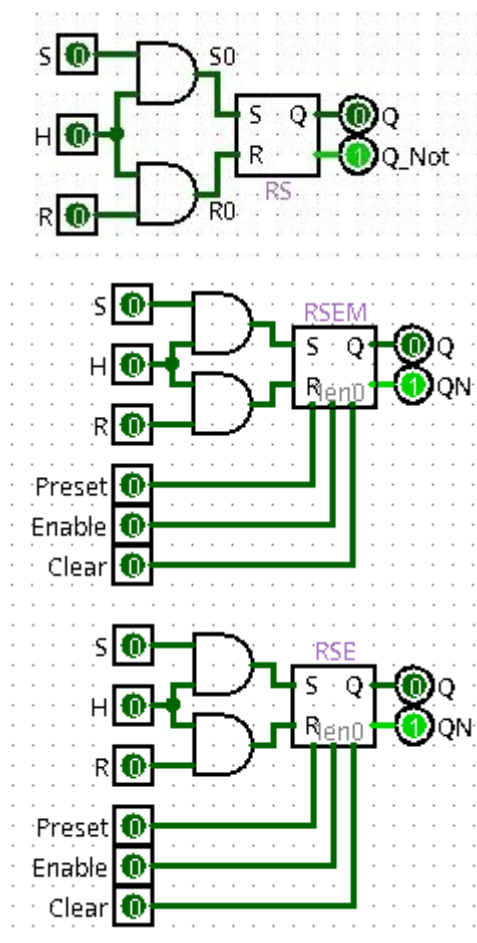
	Clear	Enable	Preset	ClearOut	EnableOut	PresetOut	
0	0	0	0	0	0	0	
1	0	0	1	0	0	1	PresetOut
2	0	1	0	0	1	0	EnableOut
3	0	1	1	0	0	1	PresetOut
4	1	0	0	1	0	0	ClearOut
5	1	0	1	1	0	0	ClearOut
6	1	1	0	1	0	0	ClearOut
7	1	1	1	1	0	0	ClearOut

2-4) Bascule : (RSE => RS Enable)



3) Bascule RSH :

3-1) Bascule : (High Level [Niveau Haut])



Algèbre de boole

$$S0 = S \cdot H$$

$$R0 = R \cdot H$$

$$Q = Q_{(n-1)} \cdot (\bar{R} + \bar{H}) + S \cdot H$$

$$\bar{Q} = \bar{Q}_{(n-1)} \cdot (\bar{S} + \bar{H}) + R \cdot H$$

$$Q = Q_{(n-1)} \cdot \bar{R}0 + S0 \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{S}0 + R0$$

$$Q = Q_{(n-1)} \cdot \bar{R} \cdot \bar{H} + S \cdot H \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{S} \cdot \bar{H} + R \cdot H$$

$$Q = Q_{(n-1)} \cdot (\bar{R} + \bar{H}) + S \cdot H \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot (\bar{S} + \bar{H}) + R \cdot H$$

S	R	H	Q	$\bar{Q}$	remarque
X	X	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	0	1	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	1	0	1	mise à 0
1	0	1	1	0	mise à 1
1	1	1	0	0	état interdit

H=0

$$Q = Q_{(n-1)} \cdot (\bar{R} + \bar{0}) + S \cdot 0 \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot (\bar{S} + \bar{0}) + R \cdot 0$$

$$Q = Q_{(n-1)} \cdot (\bar{R} + 1) + S \cdot 0 \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot (\bar{S} + 1) + R \cdot 0$$

$$Q = Q_{(n-1)} \cdot 1 \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot 1$$

$$Q = Q_{(n-1)} \quad \bar{Q} = \bar{Q}_{(n-1)}$$

$$Q = Q_{(n-1)}$$

$$\bar{Q} = \bar{Q}_{(n-1)}$$

Mémoire

H=1

$$Q = Q_{(n-1)} \cdot (\bar{R} + \bar{1}) + S \cdot 1 \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot (\bar{S} + \bar{1}) + R \cdot 1$$

$$Q = Q_{(n-1)} \cdot (\bar{R} + 0) + S \cdot 1 \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot (\bar{S} + 0) + R \cdot 1$$

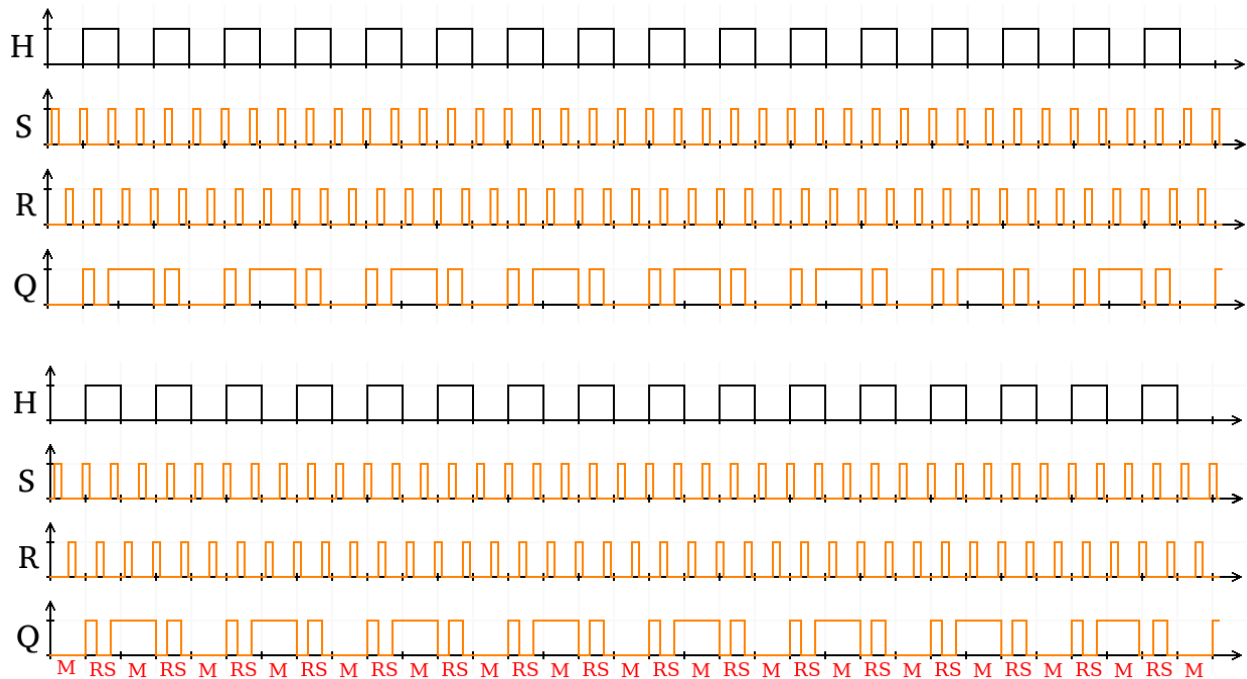
$$Q = Q_{(n-1)} \cdot \bar{R} + S \quad \bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{S} + R$$

$$Q = Q_{(n-1)} \cdot \bar{R} + S$$

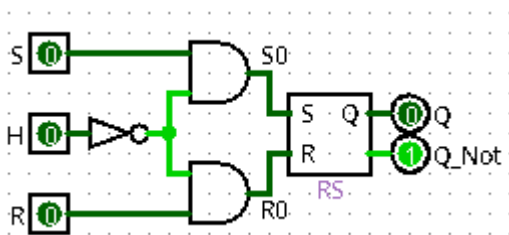
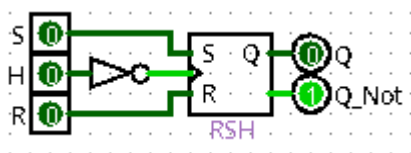
$$\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{S} + R$$

Bascule RS

3-2) diagramme de temps :



3-3) Logisim : (RSHB => RSH Bas, Low Level [Niveau Bas])



$$Q = Q_{(n-1)} \cdot (\bar{R} + H) + S \cdot \bar{H}$$

$$\bar{Q} = \bar{Q}_{(n-1)} \cdot (\bar{S} + H) + R \cdot \bar{H}$$

H=1

$$Q = Q_{(n-1)}$$

$$\bar{Q} = \bar{Q}_{(n-1)}$$

Mémoire

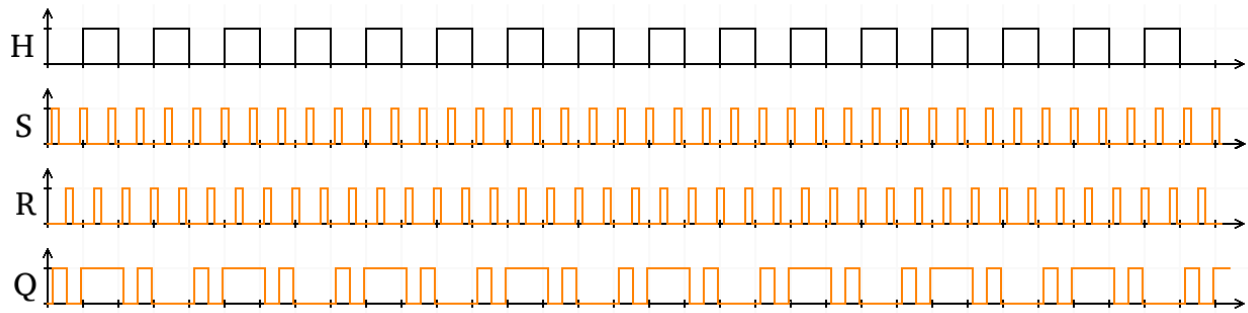
H=0

$$Q = Q_{(n-1)} \cdot \bar{R} + S$$

$$\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{S} + R$$

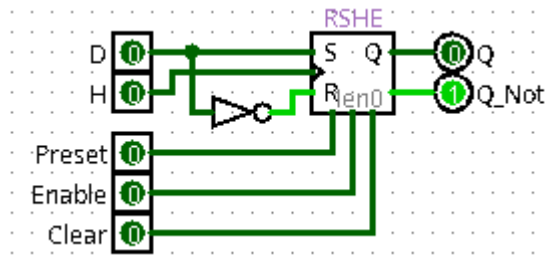
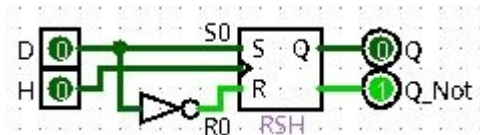
Bascule RS

S	R	H	Q	$\bar{Q}$	remarque
X	X	1	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	0	1	mise à 0
1	0	0	1	0	mise à 1
1	1	0	0	0	état interdit



4) Bascule D : (D avec RSH)

4-1) Bascule :



$$Q = Q_{(n-1)} \cdot \overline{H} + D \cdot (Q_{(n-1)} + H)$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{H} + \overline{D} \cdot (\overline{Q_{(n-1)}} + H)$$

D	H	Q	$\overline{Q}$	remarque
X	0	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	1	0	1	mise à 0
1	1	1	0	mise à 1

Algèbre de boole

$$S0 = D$$

$$R0 = \overline{D}$$

$$Q = Q_{(n-1)} \cdot (\overline{R0} + \overline{H}) + S0 \cdot H$$

$$Q = Q_{(n-1)} \cdot (\overline{\overline{D}} + \overline{H}) + D \cdot H$$

$$Q = Q_{(n-1)} \cdot (D + \overline{H}) + D \cdot H$$

$$Q = Q_{(n-1)} \cdot \overline{H} + Q_{(n-1)} \cdot D + D \cdot H$$

$$Q = Q_{(n-1)} \cdot \overline{H} + D \cdot (Q_{(n-1)} + H)$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{S0} + \overline{H}) + R0 \cdot H$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{D} + \overline{H}) + \overline{D} \cdot H$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{D} + \overline{Q_{(n-1)}} \cdot \overline{H} + \overline{D} \cdot H$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{H} + \overline{D} \cdot (\overline{Q_{(n-1)}} + H)$$

H=0

$$Q = Q_{(n-1)} \cdot \bar{0} + D \cdot (Q_{(n-1)} + 0)$$

$$Q = Q_{(n-1)} \cdot 1 + D \cdot (Q_{(n-1)} + 0)$$

$$Q = Q_{(n-1)} + D \cdot Q_{(n-1)}$$

$$Q = Q_{(n-1)} \cdot (1 + D)$$

$$Q = Q_{(n-1)} \cdot 1$$

$$Q = Q_{(n-1)}$$

$$\bar{Q} = \overline{Q_{(n-1)}} \cdot \bar{0} + \bar{D} \cdot (\overline{Q_{(n-1)}} + 0)$$

$$\bar{Q} = \overline{Q_{(n-1)}} \cdot 1 + \bar{D} \cdot (\overline{Q_{(n-1)}} + 0)$$

$$\bar{Q} = \overline{Q_{(n-1)}} + \bar{D} \cdot \overline{Q_{(n-1)}}$$

$$\bar{Q} = \overline{Q_{(n-1)}} \cdot (\bar{D} + 1)$$

$$\bar{Q} = \overline{Q_{(n-1)}} \cdot 1$$

$$\bar{Q} = \overline{Q_{(n-1)}}$$

$$Q = Q_{(n-1)}$$

$$\bar{Q} = \overline{Q_{(n-1)}}$$

H=1

$$Q = Q_{(n-1)} \cdot \bar{1} + D \cdot (Q_{(n-1)} + 1)$$

$$Q = Q_{(n-1)} \cdot 0 + D \cdot (Q_{(n-1)} + 1)$$

$$Q = 0 + D \cdot 1$$

$$Q = D$$

$$\bar{Q} = \overline{Q_{(n-1)}} \cdot \bar{1} + \bar{D} \cdot (\overline{Q_{(n-1)}} + 1)$$

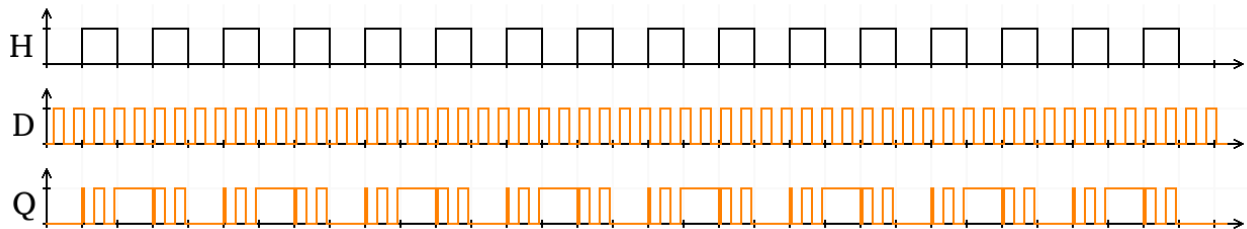
$$\bar{Q} = \overline{Q_{(n-1)}} \cdot 0 + \bar{D} \cdot (\overline{Q_{(n-1)}} + 1)$$

$$\bar{Q} = 0 + \bar{D} \cdot 1$$

$$\bar{Q} = \bar{D}$$

$$Q = D$$

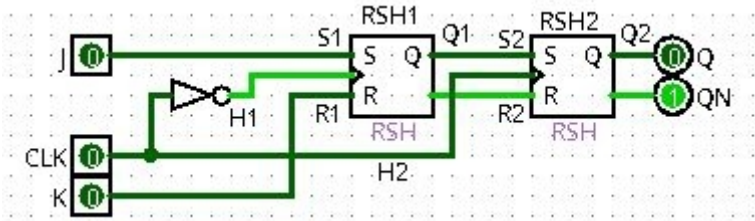
$$\bar{Q} = \bar{D}$$

4-2) diagramme de temps :

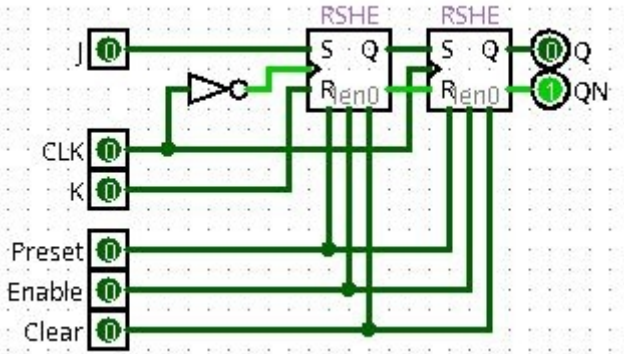
5) Bascule JK : (JKM => JK Montant)

5-1) Bascule :

Rising Edge [Front montant]



S	R	CLK	Q	Q̄	remarque
X	X	X	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	0	↑	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	1	↑	0	1	mise à 0
1	0	↑	1	0	mise à 1
1	1	↑	0	0	état interdit



5-2) RSH1 :

Algèbre de boole	CLK=0	CLK=1
$H1 = \overline{CLK}$	$Q1 = Q1_{(n-1)} \cdot (\overline{K} + 0) + J \cdot \overline{0}$	$Q1 = Q1_{(n-1)} \cdot (\overline{K} + 1) + J \cdot \overline{1}$
$Q1 = Q1_{(n-1)} \cdot (\overline{R1} + \overline{H1}) + S1 \cdot H1$	$Q1 = Q1_{(n-1)} \cdot \overline{K} + J \cdot 1$	$Q1 = Q1_{(n-1)} \cdot 1 + J \cdot 0$
$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{S1} + \overline{H1}) + R1 \cdot H1$	$Q1 = Q1_{(n-1)} \cdot \overline{K} + J$	$Q1 = Q1_{(n-1)}$
$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{CLK}) + J \cdot \overline{CLK}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + 0) + K \cdot \overline{0}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + 1) + K \cdot \overline{1}$
$Q1 = Q1_{(n-1)} \cdot (\overline{K} + CLK) + J \cdot \overline{CLK}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K \cdot 1$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot 1 + K \cdot 0$
$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{CLK}) + K \cdot \overline{CLK}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K$	$\overline{Q1} = \overline{Q1_{(n-1)}}$
$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + CLK) + K \cdot \overline{CLK}$	$Q1 = Q1_{(n-1)} \cdot \overline{K} + J$	$Q1 = Q1_{(n-1)}$
$Q1 = Q1_{(n-1)} \cdot (\overline{K} + CLK) + J \cdot \overline{CLK}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K$	$\overline{Q1} = \overline{Q1_{(n-1)}}$
$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + CLK) + K \cdot \overline{CLK}$	Bascule RS (S=J et R=K)	Mémoire

5-3) RSH2 :

Algèbre de boole	CLK=0	CLK=1
$H1 = CLK$	$Q2 = Q2_{(n-1)} \cdot (Q1 + \overline{0}) + Q1 \cdot 0$	$Q2 = Q2_{(n-1)} \cdot (Q1 + \overline{1}) + Q1 \cdot 1$
$Q2 = Q2_{(n-1)} \cdot (\overline{R2} + \overline{H2}) + S2 \cdot H2$	$Q2 = Q2_{(n-1)} \cdot (Q1 + 1)$	$Q2 = Q2_{(n-1)} \cdot (Q1 + 0) + Q1$
$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{S2} + \overline{H2}) + R2 \cdot H2$	$Q2 = Q2_{(n-1)} \cdot 1$	$Q2 = Q2_{(n-1)} \cdot Q1 + Q1$
$Q2 = Q2_{(n-1)} \cdot (\overline{Q1} + \overline{CLK}) + Q1 \cdot CLK$	$Q2 = Q2_{(n-1)}$	$Q2 = (Q2_{(n-1)} + 1) \cdot Q1$
$Q2 = Q2_{(n-1)} \cdot (Q1 + \overline{CLK}) + Q1 \cdot CLK$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{0}) + \overline{Q1} \cdot 0$	$Q2 = Q1$
$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{CLK}) + \overline{Q1} \cdot CLK$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 1)$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{1}) + \overline{Q1} \cdot 1$
$Q2 = Q2_{(n-1)} \cdot (Q1 + \overline{CLK}) + Q1 \cdot CLK$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot 1$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 0) + \overline{Q1}$
$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{CLK}) + \overline{Q1} \cdot CLK$	$\overline{Q2} = \overline{Q2_{(n-1)}}$	$\overline{Q2} = Q2_{(n-1)} \cdot \overline{Q1} + \overline{Q1}$
	$Q2 = Q2_{(n-1)}$	$\overline{Q2} = (\overline{Q2_{(n-1)}} + 1) \cdot \overline{Q1}$
	$\overline{Q2} = \overline{Q2_{(n-1)}}$	$\overline{Q2} = \overline{Q1}$
	Mémoire	$Q2 = Q1$
		$\overline{Q2} = \overline{Q1}$
		Bascule D (D=Q1)

5-4) diagramme de temps pour RSH1 et RSH2 :

CLK=0

$$Q1 = Q1_{(n-1)} \cdot \bar{K} + J$$

$$\bar{Q1} = \bar{Q1}_{(n-1)} \cdot \bar{J} + K$$

$$Q2 = Q2_{(n-1)}$$

$$\bar{Q2} = \bar{Q2}_{(n-1)}$$

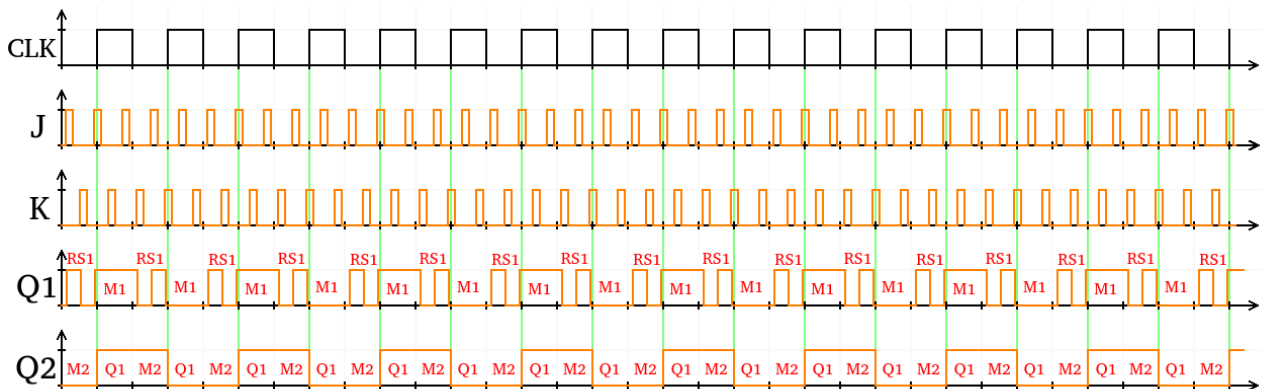
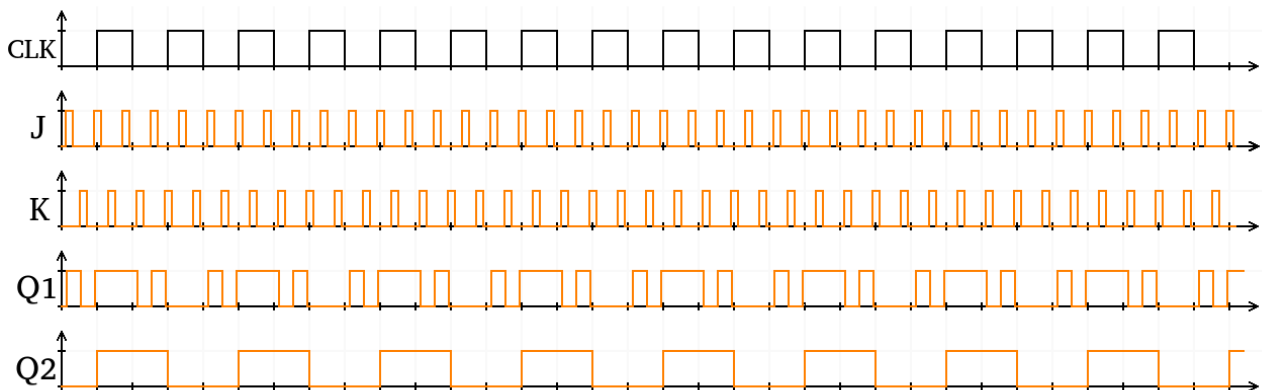
CLK=1

$$Q2 = Q1$$

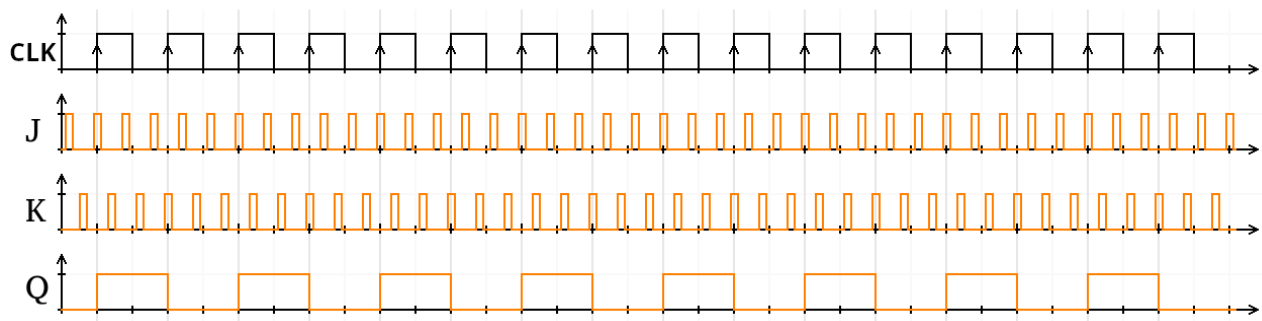
$$\bar{Q2} = \bar{Q1}$$

$$Q1 = Q1_{(n-1)}$$

$$\bar{Q1} = \bar{Q1}_{(n-1)}$$



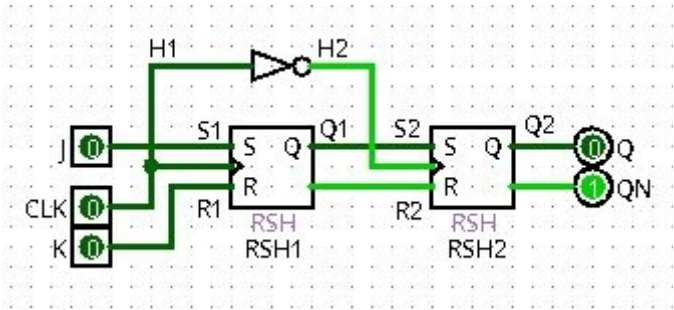
5-5) diagramme de temps pour JK :



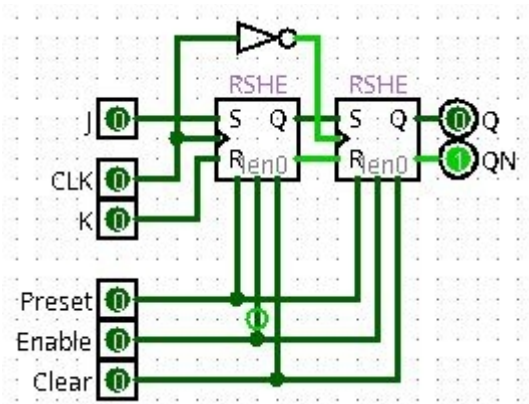
6) Bascule JK : (JKD => JK Descendant)

6-1) Bascule :

Rising Edge [Front montant]



S	R	<u>CLK</u>	Q	$\overline{Q}$	remarque
X	X	X	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	0	↓	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	1	↓	0	1	mise à 0
1	0	↓	1	0	mise à 1
1	1	↓	0	0	état interdit



6-2) RSH1 :**Algèbre de boole**

$$H1 = CLK$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{R1} + \overline{H1}) + S1 \cdot H1$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{S1} + \overline{H1}) + R1 \cdot H1$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{CLK}) + J \cdot CLK$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{CLK}) + K \cdot CLK$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{CLK}) + J \cdot CLK$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{CLK}) + K \cdot CLK$$

CLK=0

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{0}) + J \cdot 0$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + 1)$$

$$Q1 = Q1_{(n-1)} \cdot 1$$

$$Q1 = Q1_{(n-1)}$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{0}) + K \cdot 0$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + 1)$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot 1$$

$$\overline{Q1} = \overline{Q1_{(n-1)}}$$

$$Q1 = Q1_{(n-1)}$$

$$\overline{Q1} = \overline{Q1_{(n-1)}}$$

Mémoire

CLK=1

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{1}) + J \cdot 1$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + 0) + J$$

$$Q1 = Q1_{(n-1)} \cdot \overline{K} + J$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{1}) + K \cdot 1$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + 0) + K$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K$$

$$Q1 = Q1_{(n-1)} \cdot \overline{K} + J$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K$$

Bascule RS (S=J et R=K)

6-3) RSH2 :

$$H1 = \overline{CLK}$$

$$Q2 = Q2_{(n-1)} \cdot (\overline{R2} + \overline{H2}) + S2 \cdot H2$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{S2} + \overline{H2}) + R2 \cdot H2$$

$$Q2 = Q2_{(n-1)} \cdot (\overline{Q1} + \overline{CLK}) + Q1 \cdot \overline{CLK}$$

$$Q2 = Q2_{(n-1)} \cdot (Q1 + CLK) + Q1 \cdot \overline{CLK}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{CLK}) + \overline{Q1} \cdot \overline{CLK}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + CLK) + \overline{Q1} \cdot CLK$$

$$Q2 = Q2_{(n-1)} \cdot (Q1 + CLK) + Q1 \cdot \overline{CLK}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + CLK) + \overline{Q1} \cdot \overline{CLK}$$

CLK=0

$$Q2 = Q2_{(n-1)} \cdot (Q1 + 0) + Q1 \cdot \overline{0}$$

$$Q2 = Q2_{(n-1)} \cdot Q1 + Q1 \cdot 1$$

$$Q2 = Q2_{(n-1)} \cdot Q1 + Q1$$

$$Q2 = (Q2_{(n-1)} + 1) \cdot Q1$$

$$Q2 = 1 \cdot Q1$$

$$Q2 = Q1$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 0) + \overline{Q1} \cdot \overline{0}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot \overline{Q1} + \overline{Q1} \cdot 1$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot \overline{Q1} + \overline{Q1}$$

$$\overline{Q2} = (\overline{Q2_{(n-1)}} + 1) \cdot \overline{Q1}$$

$$\overline{Q2} = 1 \cdot \overline{Q1}$$

$$\overline{Q2} = \overline{Q1}$$

$$Q2 = Q1$$

$$\overline{Q2} = \overline{Q1}$$

Bascule D (D=Q1)

CLK=1

$$Q2 = Q2_{(n-1)} \cdot (Q1 + 1) + Q1 \cdot \overline{1}$$

$$Q2 = Q2_{(n-1)} \cdot (Q1 + 1) + Q1 \cdot 0$$

$$Q2 = Q2_{(n-1)} \cdot 1 + 0$$

$$Q2 = Q2_{(n-1)}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 1) + \overline{Q1} \cdot \overline{1}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 1) + \overline{Q1} \cdot 0$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot 1 + 0$$

$$\overline{Q2} = \overline{Q2_{(n-1)}}$$

$$Q2 = Q2_{(n-1)}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}}$$

Mémoire

6-4) diagramme de temps pour RSH1 et RSH2 :

CLK=0

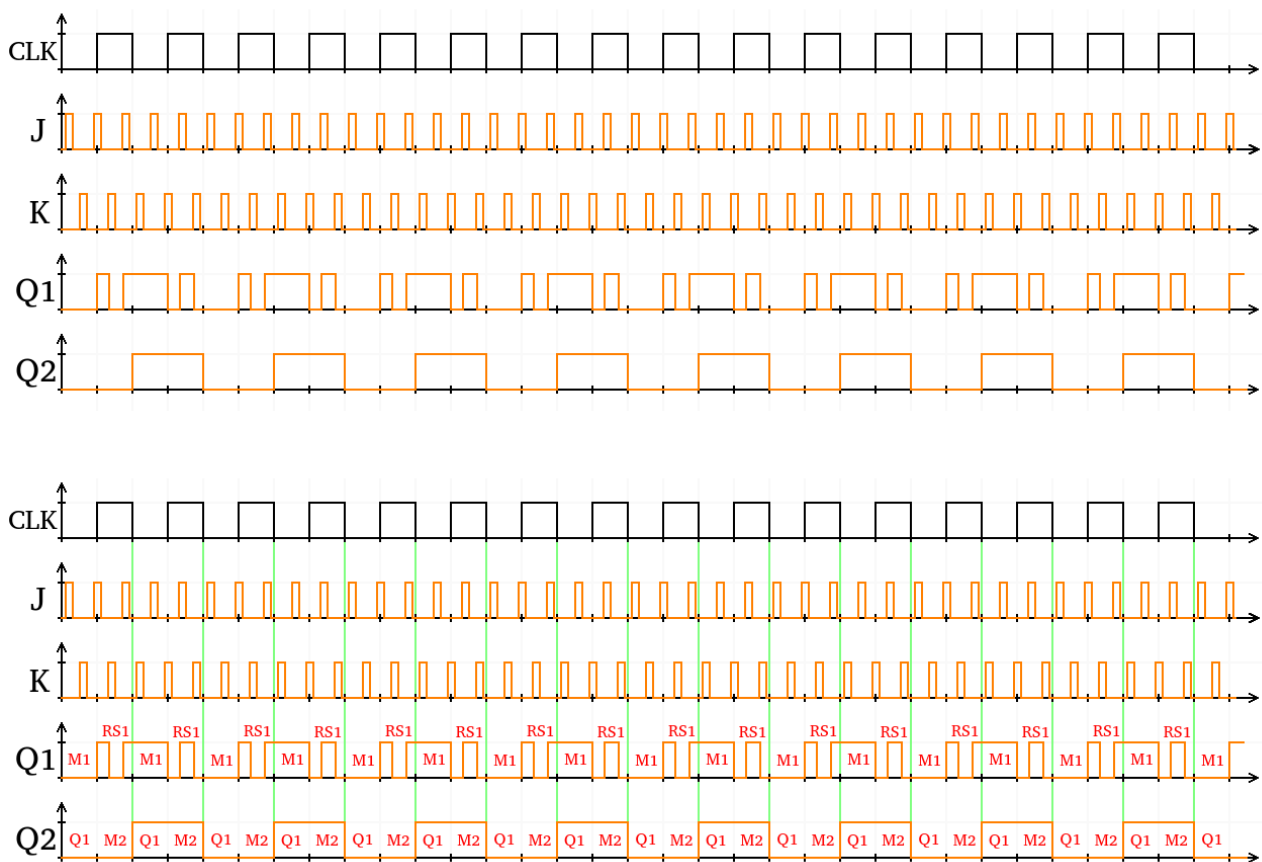
$$\frac{Q2}{Q2} = \frac{Q1}{Q1}$$

$$\frac{Q1}{Q1} = \frac{Q1_{(n-1)}}{Q1_{(n-1)}}$$

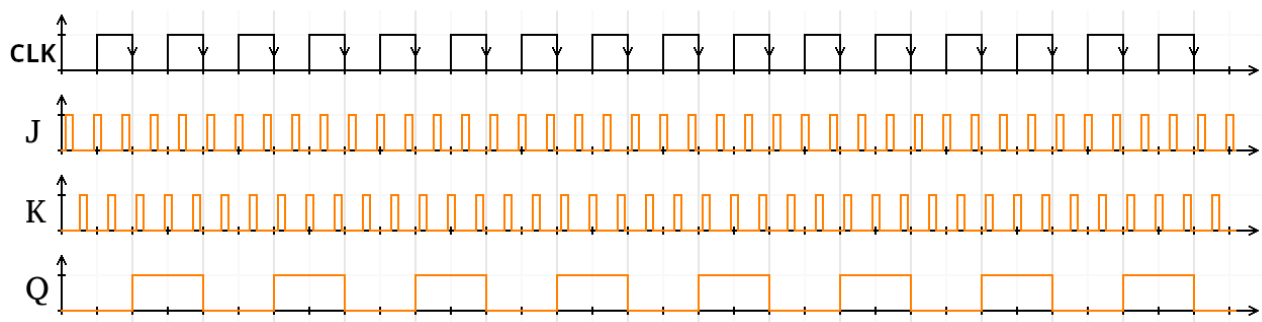
CLK=1 :

$$\frac{Q1}{Q1} = \frac{Q1_{(n-1)} \cdot \bar{K} + J}{Q1_{(n-1)} \cdot \bar{J} + K}$$

$$\frac{Q2}{Q2} = \frac{Q2_{(n-1)}}{Q2_{(n-1)}}$$

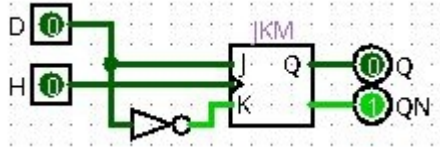
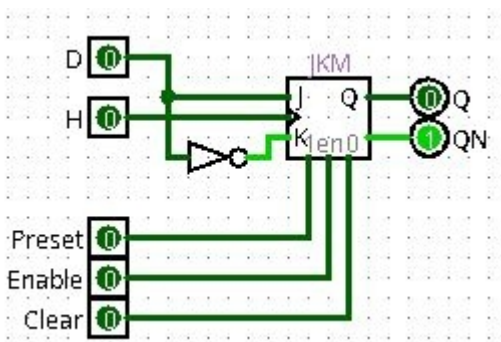


6-5) diagramme de temps pour JK :

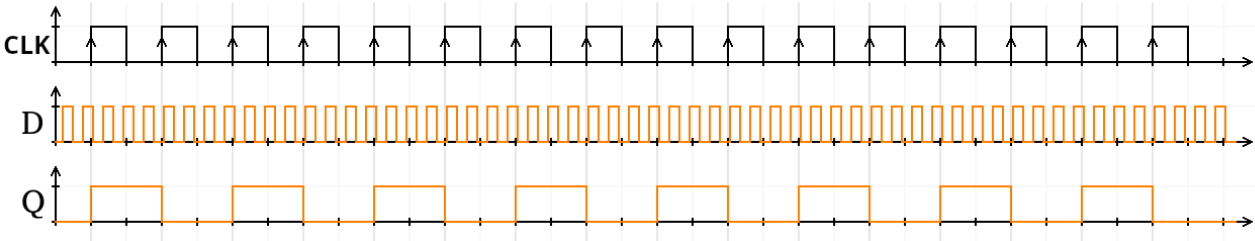


7) Bascule D : (DM => D Montant)

Rising Edge [Front montant]

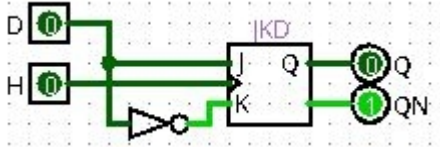
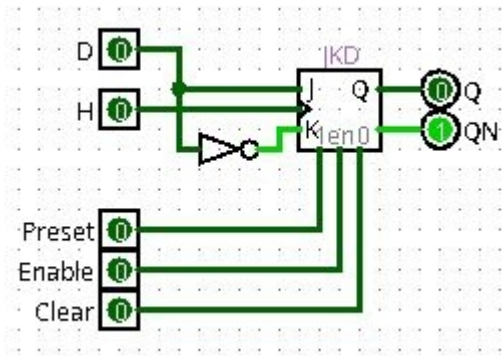


D	CLK	Q	Q̄	remarque
X	X	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	↑	0	1	mise à 0
1	↑	1	0	mise à 1



8) Bascule D : (DD => D Descendant)

Falling Edge [Front descendant]



D	CLK	Q	Q̄	remarque
X	X	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	↓	0	1	mise à 0
1	↓	1	0	mise à 1

